Ethan B. Chen

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Education

Vanderbilt University **Expected May 2030**
B.S. in Computer Science / Second Major: Human & Organizational Development *Nashville, TN*
Honors: National Merit Finalist | Presidential Scholar Semifinalist

Experience

Main Developer, City Settlers **Sep. 2026 – Present**
Change++ — Civic Design Center *Vanderbilt University*
- Re-architected City Settlers, a multiplayer civic-engagement game for Nashville's Civic Design Center, into an idle city builder with shared-resource policy mechanics; implemented **13.9K lines across 78 files** in a draft PR.
- Engineered deterministic TypeScript game rules and server-authoritative Node/Socket.IO multiplayer; added **70 automated tests** and load-tested **30 simulated students** at **1 ms p50 / 13 ms p95** server latency.

Full-Stack Software Engineering Intern **Sep. 2026 – Present**
Lawly *Remote*
- Built Lawly's initial **Playwright E2E regression framework**, automating six authentication/session behaviors and adding contract validation against the production sign-in handler.

Web Development Intern **Sep. 2026 – Present**
Alterea, Inc. — Agents of Influence *Remote*
- Investigated **6** React/TypeScript dashboard issues and opened **5 bug-fix PRs** across onboarding, classroom routing, CSV import, pathway UI, and student selection; validated fixes against **493 passing tests** plus Playwright, lint, typecheck, and build checks.

Founder & Web Developer **Sep. 2024 – Present**
Wifi Concierge *Westlake, OH*
- Founded a community technology service serving **70+** seniors and underserved families and raised **$1,500+** through technical services and device resale.

NextGen Radiology Department Intern **Jun. 2025 – Aug. 2025**
Cleveland Clinic *Cleveland, OH*
- Worked with clinical AI, machine learning, image processing, and medical imaging under Po-Hao Chen, MD, MBA, Vice Chair of AI, Diagnostics Institute.

Research & Development Intern **Jun. 2024 – Aug. 2024**
Hyland Software *Westlake, OH*
- Centralized technical resources for three Hy-Tech Club course tracks by redesigning the program website and consolidating student forms, virtual-session links, and learning materials.

Projects

The Great Prompt-Off [Live] *Next.js, TypeScript, Supabase/PostgreSQL, OpenRouter*
- Shipped a full-stack prompt-engineering competition platform on **50 synthetic knee MRI reports**, with five public attempts, a **45-report hidden final**, live leaderboards, and organizer controls.
- Implemented schema-aware scoring across six findings, persistent data, LLM evaluation, regression tests, and server-side privacy boundaries for hidden reports and answer keys.

Mosaic [Live] *React, TypeScript, Express, SQLite, Docker, Playwright*
- Shipped a full-stack image discovery/collaboration product with accounts, privacy controls, sharing, social features, recommendations, and a draggable visual Canvas.
- Built transactional Express/SQLite backend flows with permission checks, rate limiting, security headers, and automated E2E/security regression tests; deployed on Render.

TowerLogic *Python, OpenCV, PyTorch, ONNX Runtime, ADB*
- Engineered a screenshot-to-action ML pipeline combining card classification, field-object detection, normalized game state, rule masks, and learned policy scoring over a **30-value model input**.

Skills

Languages: Python, TypeScript, JavaScript, Java, Rust, SQL
Frameworks & Systems: React, Next.js, Node.js, Express, Socket.IO, React Three Fiber, three.js
Data / AI / Testing: PostgreSQL, Supabase, SQLite, PyTorch, OpenCV, ONNX Runtime, Vitest, Playwright
Developer Tools: Docker, GitHub Actions, Git/GitHub, Vercel, Render
Interests: Chess, Gaming, Piano, Music Production, Creative Coding, Interactive Web Design